

# **SYSTEM USER MANUAL FOR KCA CONNECT AGENTIC AI**

**22/05647**

**IGNATIUS LUMOSI**

**BSC INFORMATION TECHNOLOGY**

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## **Abstract**

This chapter forms part of the final year project report for KCA Connect Agentic AI, an intelligent conversational assistant developed for KCA University. The system leverages Retrieval-Augmented Generation (RAG), large language models (LLMs), and a modern web stack to enable students, faculty, and staff to access institutional information through natural language queries.

# 1. Introduction

## 1.1. Overview of the System

This user manual provides a comprehensive guide for operating the KCA Connect Agentic AI system. KCA Connect AI is an intelligent conversational assistant developed for KCA University, enabling students, faculty, and staff to access institutional information through natural language queries. The system uses Retrieval-Augmented Generation (RAG) to search and summarise information from official KCA University documents, including timetables, fee schedules, exam information, and campus announcements.

The purpose of this manual is to help users independently install, access, and use the system effectively without requiring technical support.

## 1.2. Target Users

This manual is intended for the following groups:

| User Group     | Description                                                                                   |
|----------------|-----------------------------------------------------------------------------------------------|
| Students       | KCA University students who want quick answers about timetables, fees, exams, and campus life |
| Faculty        | Lecturers and academic staff seeking administrative information                               |
| Staff          | Non-academic staff who need to look up institutional data                                     |
| Administrators | IT staff or designated super-users who manage the system, users, and knowledge base           |

## 1.3. System Requirements

Before using KCA Connect AI, ensure your environment meets the following requirements:

### *Minimum Hardware Requirements*

| Component | Requirement                                 |
|-----------|---------------------------------------------|
| Processor | Intel Core i3 / AMD Ryzen 3 (or equivalent) |
| RAM       | 4 GB minimum                                |
| Storage   | 500 MB free space (for browser cache)       |
| Display   | 1280 × 720 resolution or higher             |

### *Software Requirements*

| Component           | Requirement                                                                     |
|---------------------|---------------------------------------------------------------------------------|
| Web Browser         | Google Chrome 100+, Mozilla Firefox 100+, Microsoft Edge 100+, or Safari 14+    |
| Operating System    | Windows 10/11, macOS 11+, Ubuntu 18+, or any modern OS with a supported browser |
| Internet Connection | Stable broadband connection required                                            |

***For Local Deployment (Developers/Administrators)***

| <b>Component</b> | <b>Requirement</b>                                 |
|------------------|----------------------------------------------------|
| Python           | 3.11 or higher                                     |
| Node.js          | 18 or higher                                       |
| Docker           | Docker Engine 24+ and Docker Compose               |
| Disk Space       | At least 5 GB (for containers and embedding model) |

## 2. Getting Started

### 2.1. Installation and Setup

#### ***Option A: Running with Docker (Recommended)***

##### 1. Clone the Repository

```
git clone https://github.com/pinchez420/kca-connect-agentic-ai
cd "final year proj"
```

##### 2. Configure Environment Variables

```
cd backend
cp .env.example .env
```

Open `.env` and fill in the required API keys:

- ``GOOGLE_API_KEY`` — from [Google AI Studio](<https://makersuite.google.com/app/apikey>)
- ``GROQ_API_KEY`` — from [Groq Console](<https://console.groq.com/>)
- ``SUPABASE_URL``, ``SUPABASE_ANON_KEY`` — from your Supabase project settings
- ``QDRANT_URL`` — ``http://localhost:6333`` (default for Docker)

##### 3. Start All Services

```
docker-compose up --build
```

Services will be available at:

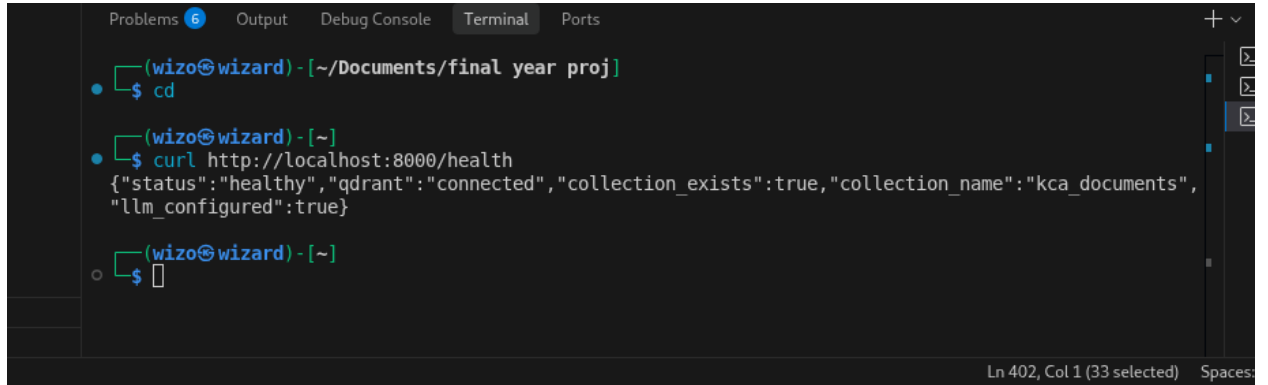
- Frontend (Chat UI): ``http://localhost:5173``
- Backend API: ``http://localhost:8000``
- Qdrant Vector DB: ``http://localhost:6333``

##### 4. Ingest Knowledge Base Documents

```
cd backend
python scripts/ingest.py
```

## 5. Verify Everything is Running

Visit <http://localhost:8000/health> — you should see a success response with all services reporting healthy status.



The screenshot shows a terminal window with the following content:

```
(wizo@wizard) - [~/Documents/final_year_proj]
$ cd
(wizo@wizard) - [~]
$ curl http://localhost:8000/health
{"status": "healthy", "qdrant": "connected", "collection_exists": true, "collection_name": "kca_documents", "llm_configured": true}
(wizo@wizard) - [~]
$
```

The terminal window has tabs for Problems, Output, Debug Console, Terminal, and Ports. The status bar at the bottom indicates "Ln 402, Col 1 (33 selected) Spaces".

### Option B: Local Development Setup

#### Frontend:

```
cd frontend
npm install
npm run dev
# Accessible at http://localhost:5173
```

#### Backend:

```
cd backend
python -m venv .venv
source .venv/bin/activate          # Windows: .venv\Scripts\activate
pip install -r requirements.txt
uvicorn main:app --reload
# API at http://localhost:8000
```



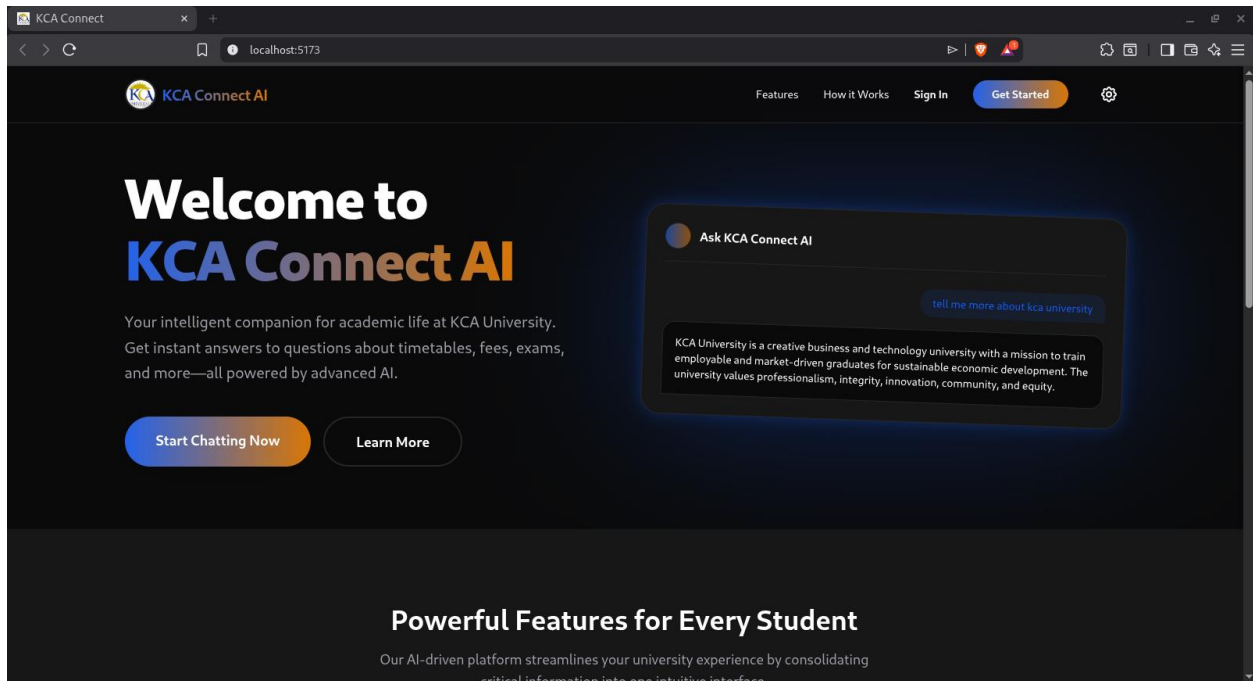
## 2.2. Accessing the System

Open your preferred web browser.

Navigate to:

Local installation: `http://localhost:5173`

You will see the KCA Connect AI Landing Page.

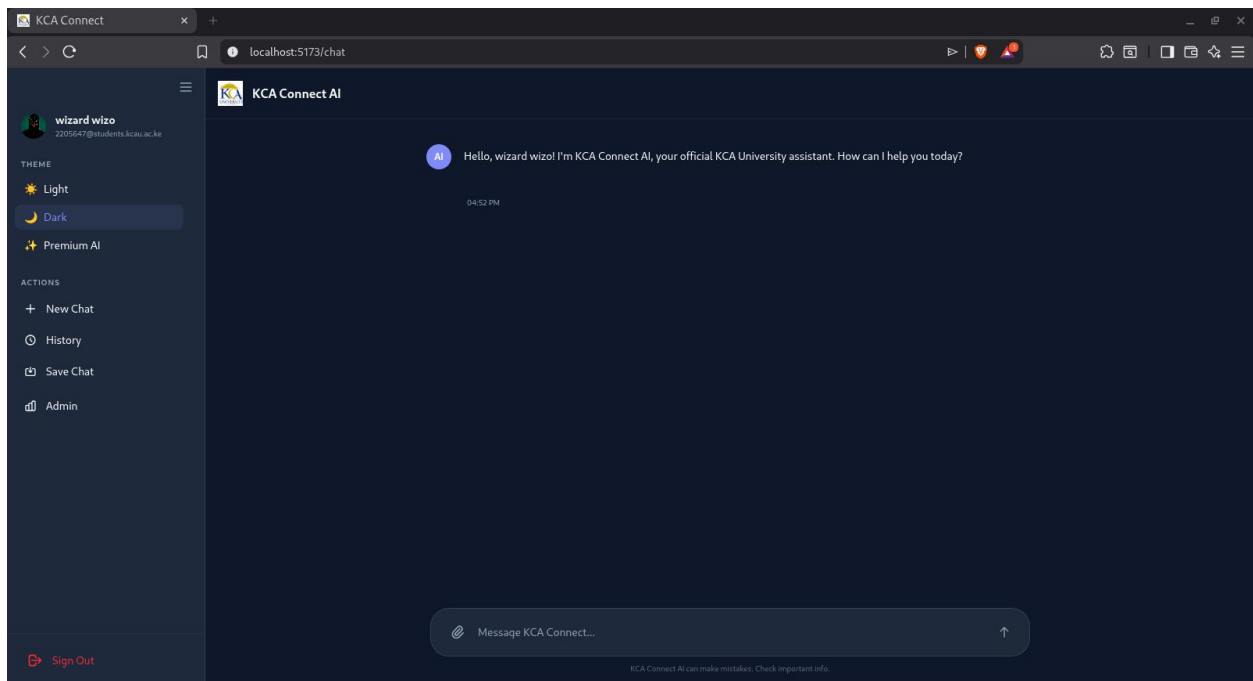


## 2.3. Interface Overview

The application has four main areas:

| Screen              | Description                                                           |
|---------------------|-----------------------------------------------------------------------|
| Landing Page        | Introduction to the system with feature highlights and sign-in access |
| Authentication Page | Sign up and sign in forms                                             |
| Chat Interface      | The main AI conversation screen                                       |
| Admin Dashboard     | System analytics and management (admin users only)                    |

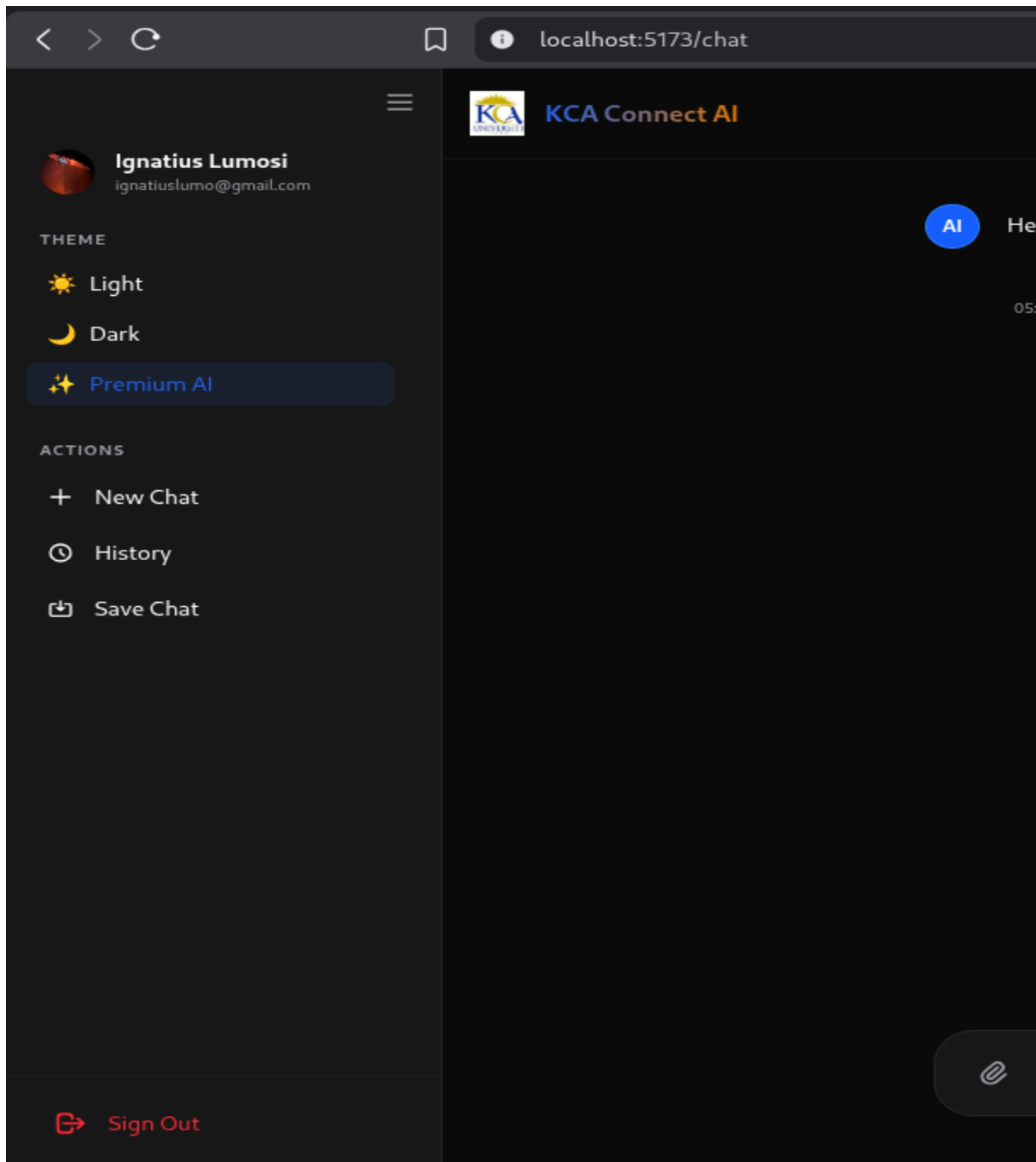
## Chat Interface Layout:



### 3. System Navigation and Core Functionalities

#### 3.1. Navigation

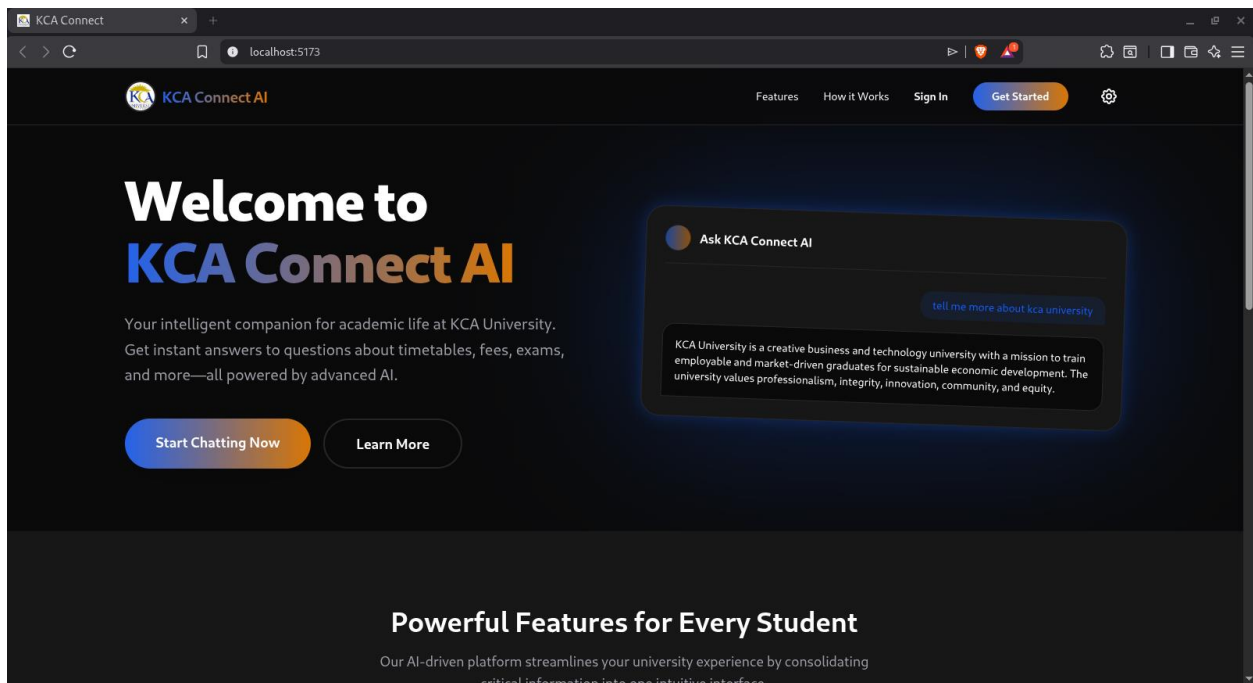
The system uses a sidebar navigation model for the authenticated chat view:

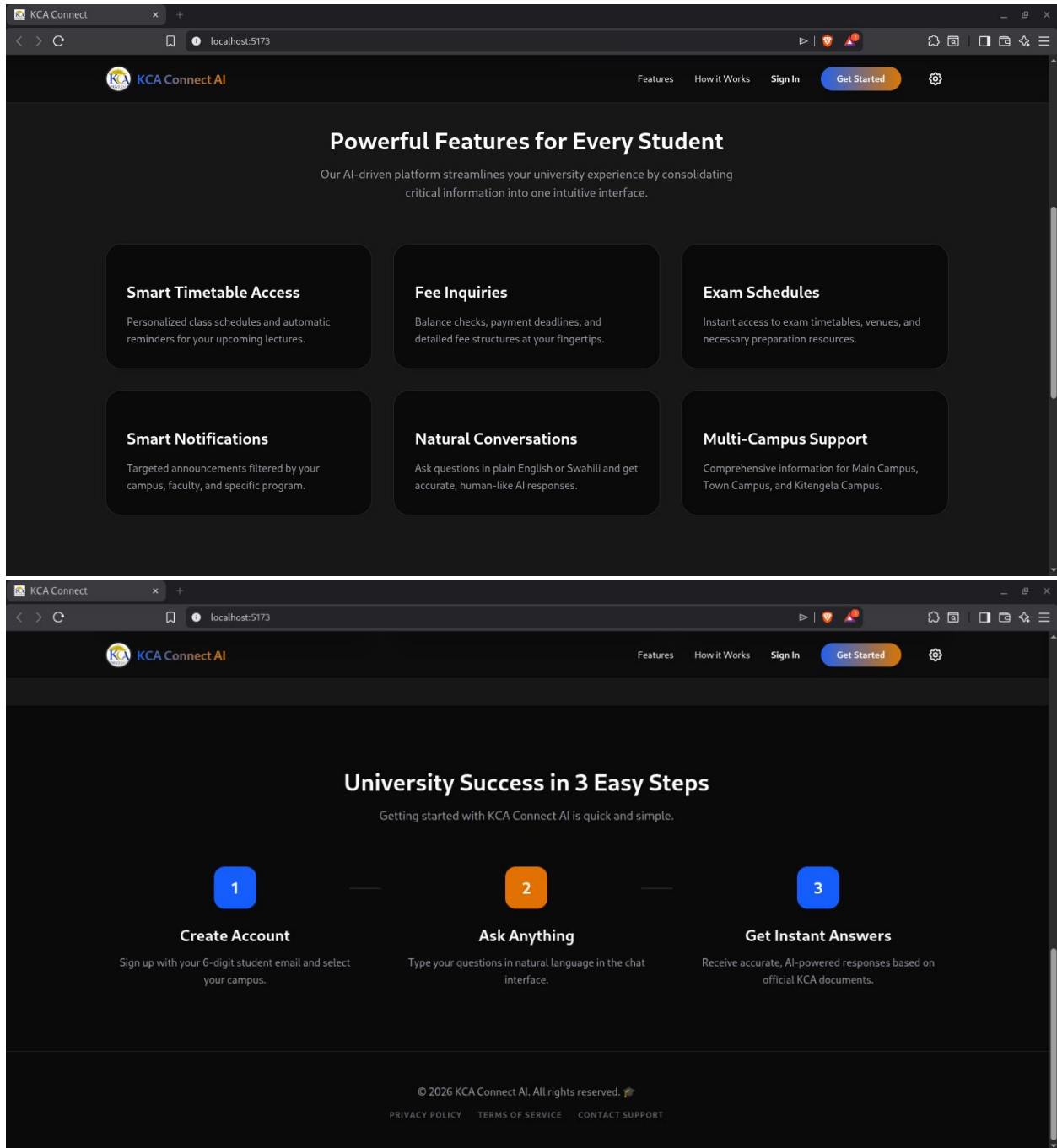


| Sidebar Item | Action                                                   |
|--------------|----------------------------------------------------------|
| New Chat     | Clears the current conversation and starts fresh         |
| Chat History | Opens a modal showing all saved conversations            |
| Save Chat    | Manually saves the current conversation                  |
| Profile      | Opens the user profile modal for viewing/editing         |
| Settings     | Opens theme and preferences settings                     |
| Sign Out     | Ends the current session and returns to the landing page |

The Landing Page navigation includes:

Features section, How It Works section, and a "Get Started" call-to-action button

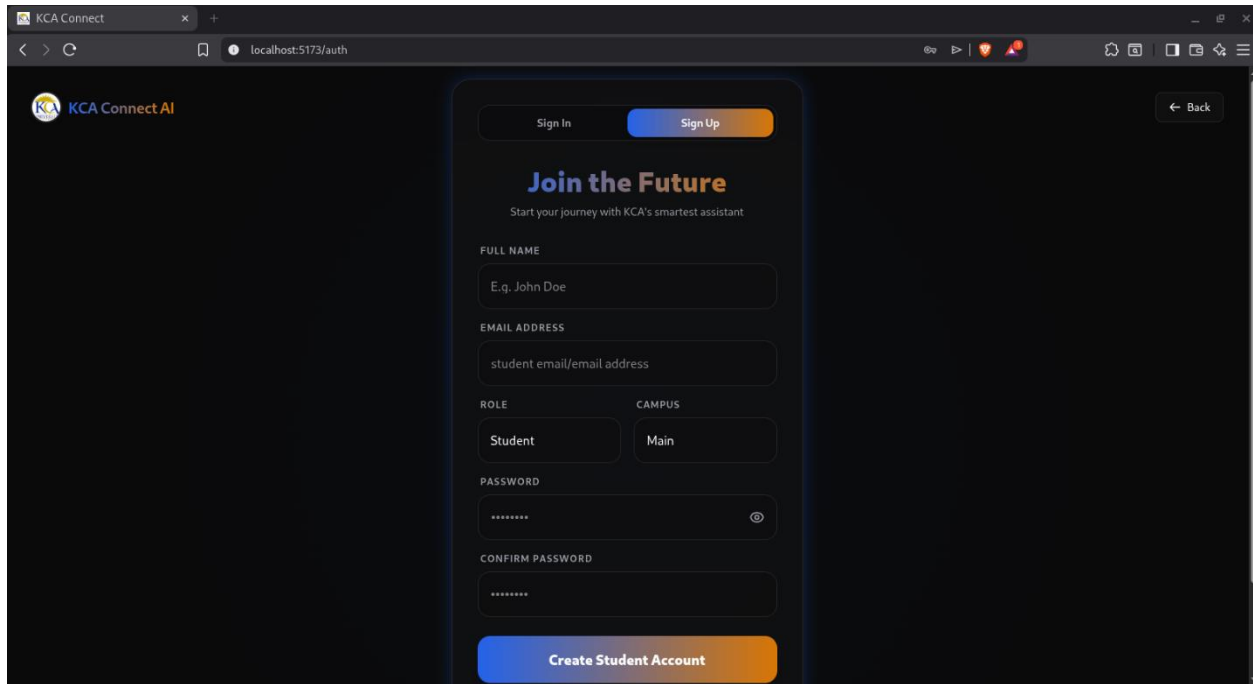




## 3.2. Core Features

### A. Creating an Account (Sign Up)

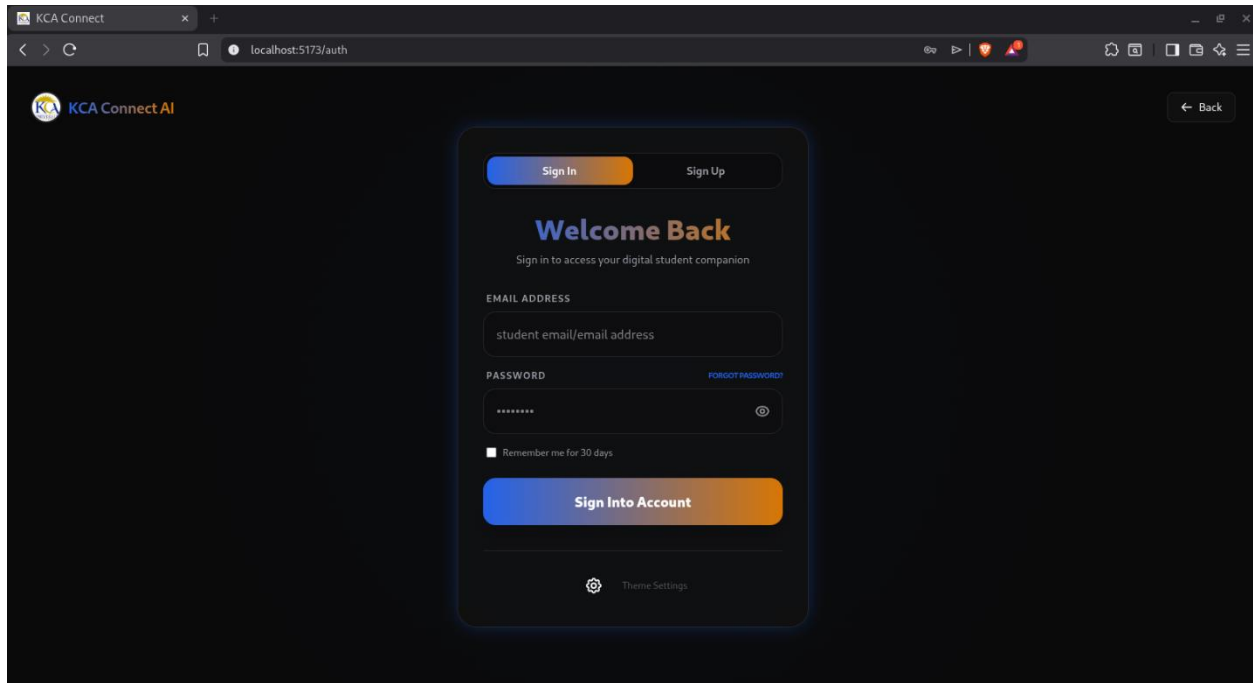
1. Navigate to the application and click "Get Started" or "Sign In".
2. Click the "Sign Up" tab.
3. Fill in all required fields:

A screenshot of a web browser displaying the KCA Connect AI sign-up page. The browser's address bar shows 'localhost:5173/auth'. The page has a dark theme. At the top left is the 'KCA Connect AI' logo. At the top right is a 'Back' button. In the center, there are two tabs: 'Sign In' and 'Sign Up', with 'Sign Up' being the active tab. Below the tabs is the heading 'Join the Future' with the subtext 'Start your journey with KCA's smartest assistant'. The form contains several input fields: 'FULL NAME' with a placeholder 'E.g. John Doe', 'EMAIL ADDRESS' with a placeholder 'student email/email address', 'ROLE' with a dropdown menu showing 'Student', 'CAMPUS' with a dropdown menu showing 'Main', 'PASSWORD' with a masked field '\*\*\*\*\*' and an eye icon, and 'CONFIRM PASSWORD' with a masked field '\*\*\*\*\*'. At the bottom of the form is a large orange button labeled 'Create Student Account'.

- Full Name — Your complete name as registered with KCA University
  - Email Address — Your student or institutional email
  - Role — Select: Student, Faculty, Staff, or Guest
  - Campus — Select: Main Campus, Town Campus, or Kitengela Campus
  - Password — Minimum 8 characters; use a combination of letters and numbers
  - Confirm Password — Re-enter to avoid typos
3. Click "Create Student Account".
  4. Check your email inbox for a confirmation email .
  5. Once confirmed, you may sign in.

## **B. Signing In**

1. Click "Sign In" from the landing page.



2. Enter your registered email address and password.
3. (Optional) Check "Remember me for 30 days" to stay signed in.
4. Click "Sign Into Account".
5. You will be redirected to the chat interface.

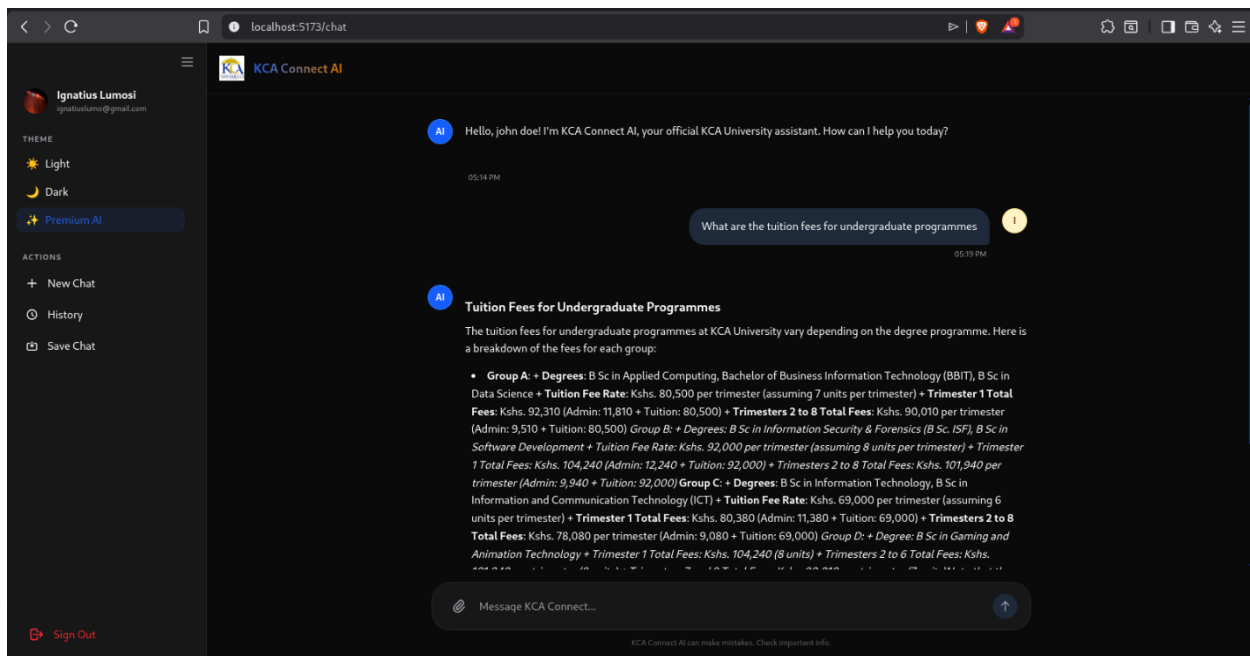
### **Forgot Password?**

6. Click "Forgot Password?" on the sign-in screen.
7. Enter your registered email.
8. Check your inbox for a password reset link.
9. Follow the link to create a new password.

### C. Using the Chat Interface

Starting a conversation:

1. Click in the text input box at the bottom of the screen.
2. Type your question in natural language. Examples:



3. Press Enter or click the Send button .
4. The AI will begin responding immediately, displaying text word-by-word.

Response features:

- Responses support bold, italic, `code`, tables, numbered lists, and headings.
- Hover over any AI message to reveal the Copy button to copy the response.
- Click the Stop button during a response to halt generation.



### **D. Managing Chat History**

Auto-save: Every conversation is automatically saved once the AI finishes responding.

Manual save:

Click "Save Chat" in the sidebar.

A success notification appears.

Viewing and loading past chats:

Click "Chat History" in the sidebar.

Browse the list of past conversations.

Click on any chat to load it into the interface.

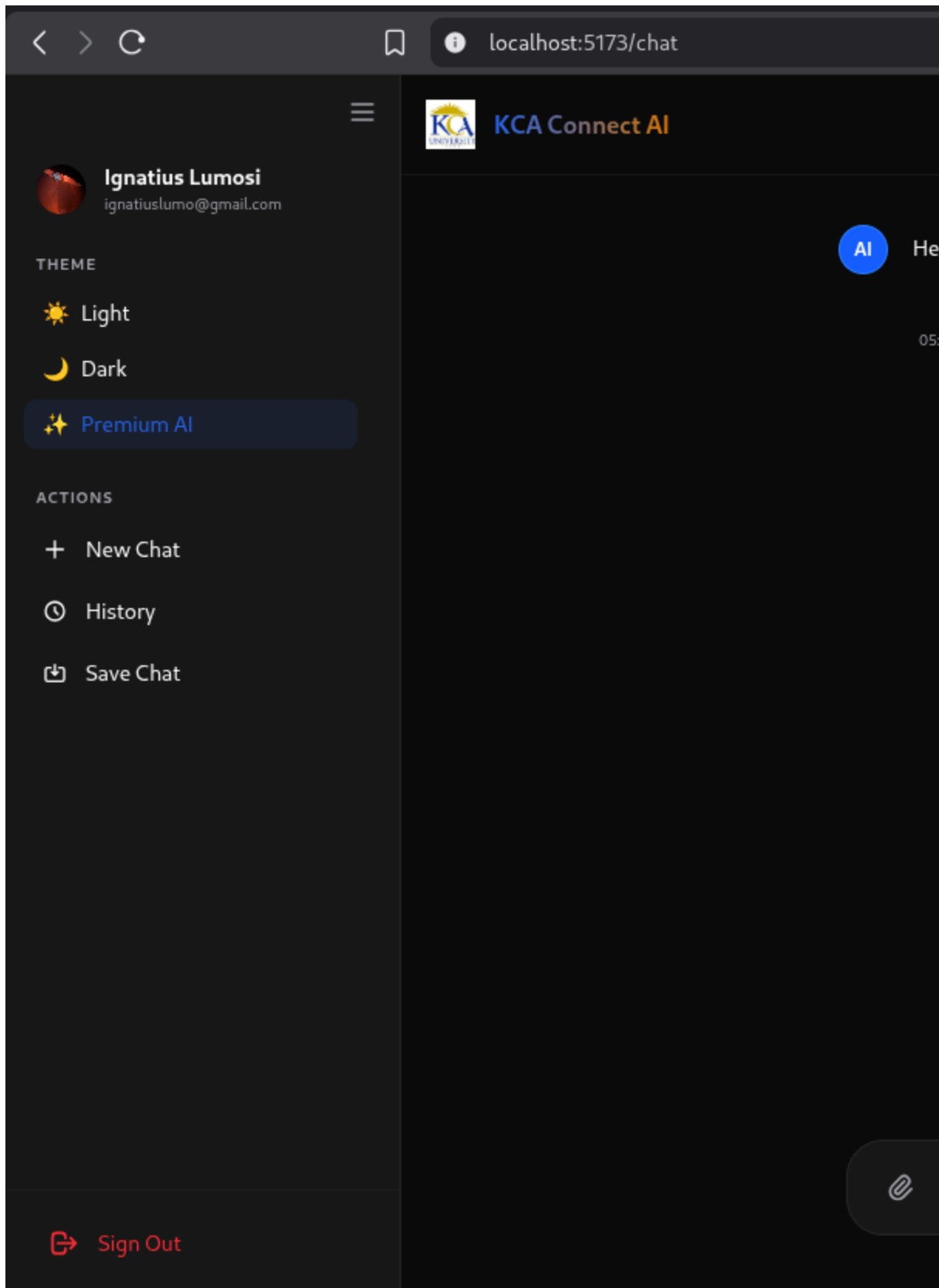
You can continue the conversation from where you left off.

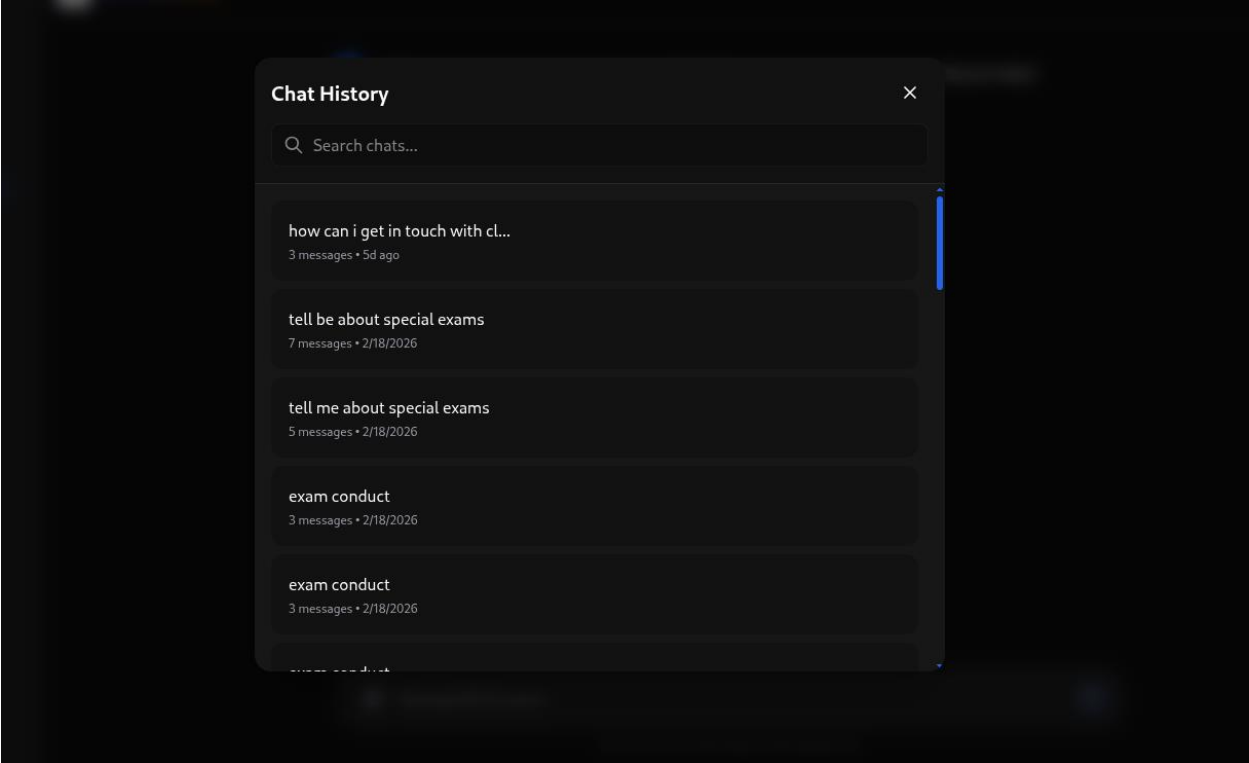
Deleting a chat:

Open Chat History.

Click the Delete icon next to the chat.

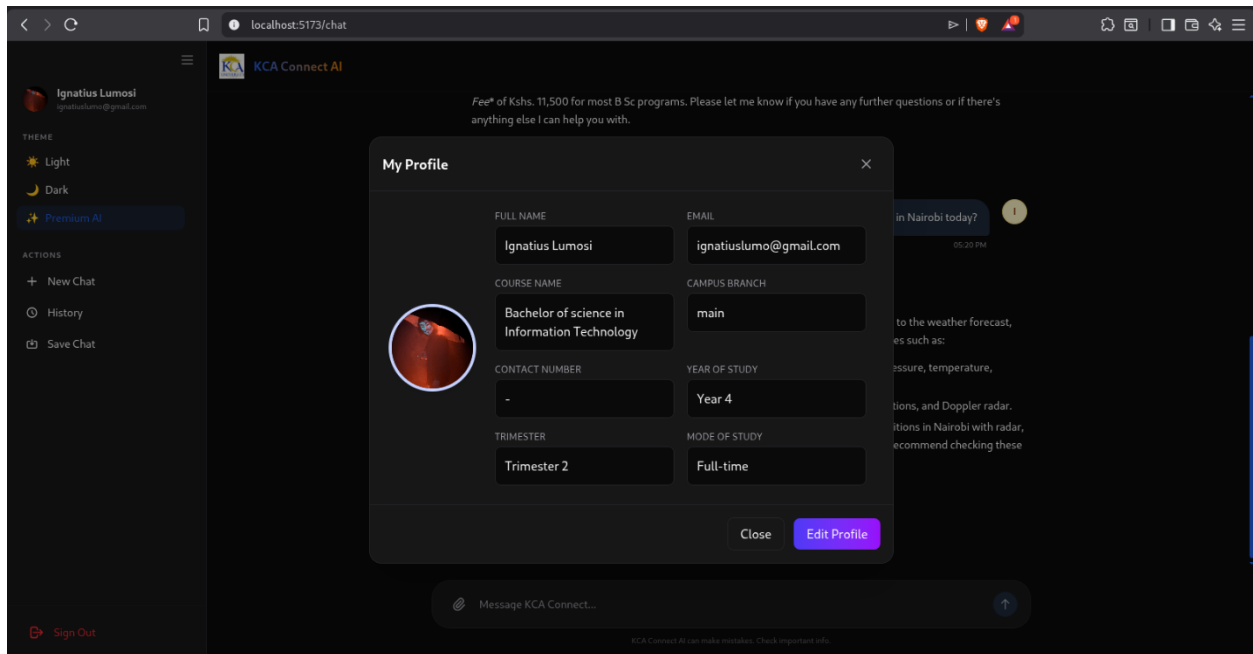
Confirm the deletion.





## E. Managing Your Profile

1. Click the Profile icon (👤) in the sidebar.
2. Your current profile information is displayed.



3. Click "Edit Profile" to modify:
  - Full Name
  - Campus Branch
  - Year of Study (for students)
  - Trimester
  - Mode of Study (Full-time / Part-time)
4. Click "Save Changes" to persist updates.

*Note: Email and Role cannot be changed after registration. Contact an administrator if changes are needed.*

Profile-aware responses: The AI uses your profile (campus, year, trimester) to provide personalised answers such as campus-specific timetables.

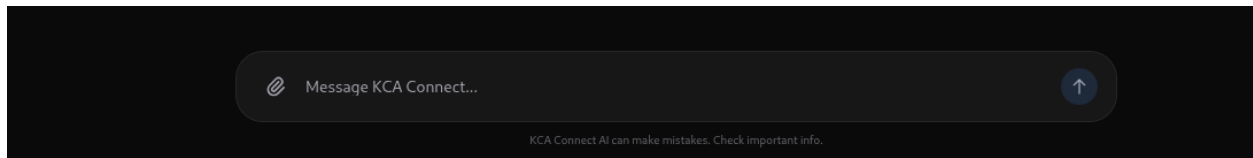
## 4. Advanced Features and Customization

### 4.1. Advanced Tools

#### **File Attachments**

You can attach documents to provide additional context for your queries:

1. Click the Attachment icon (📎) next to the message input.

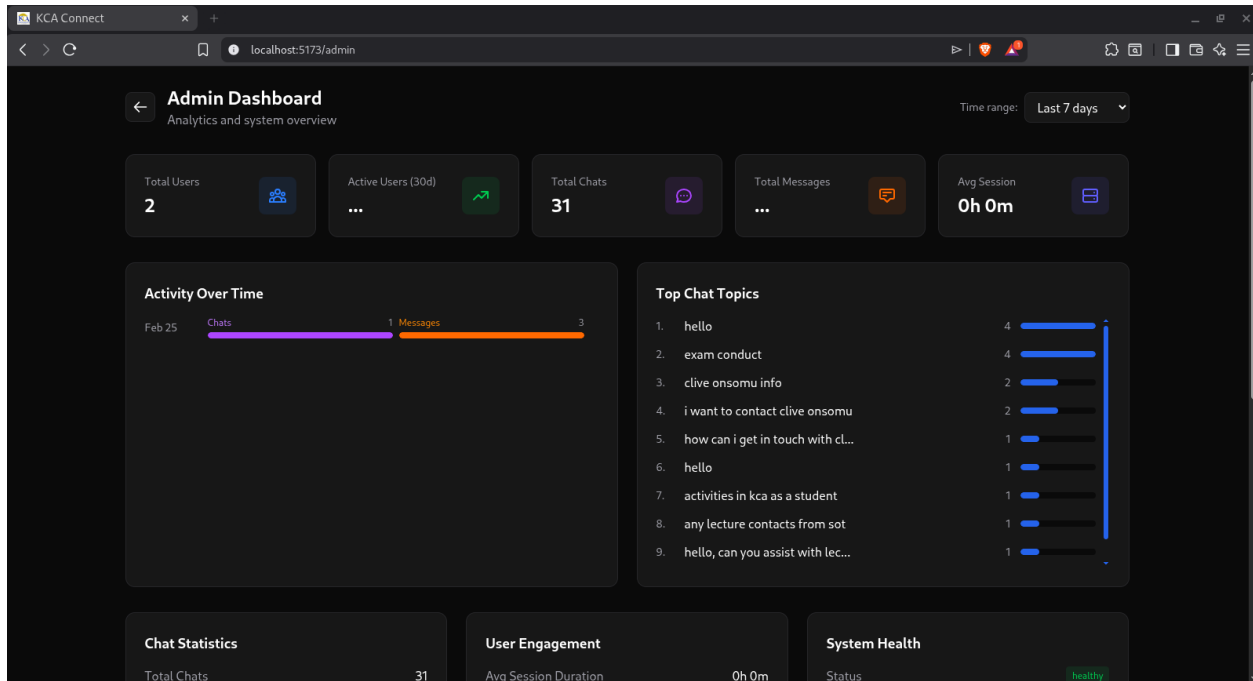


2. Select a file from your device.
  - Supported types: PDF, DOCX, TXT
  - Maximum file size: 10 MB
3. A file preview appears in the input area.
4. Type your question related to the document.
5. Send your message.
6. The AI reads the document content and incorporates it into its response.

#### **Admin Dashboard**

Only available to Administrator accounts.

Access by clicking "Admin Dashboard" in the sidebar or profile menu.



KCA Connect

localhost:5173/admin

#### Chat Statistics

|                   |     |
|-------------------|-----|
| Total Chats       | 31  |
| Saved Chats       | 0   |
| Avg Messages/Chat | 5.6 |

#### User Engagement

|                      |       |
|----------------------|-------|
| Avg Session Duration | 0h 0m |
| Avg Messages/User    | 58.3  |

#### System Health

|                    |         |
|--------------------|---------|
| Status             | healthy |
| Qdrant Collections | 1       |
| RAG Vectors        | 0       |

#### Users Management

Total: 2

Per page: 10

| Email                       | Full Name   | Campus | Chats | Admin | Created   | Actions                                                                  |
|-----------------------------|-------------|--------|-------|-------|-----------|--------------------------------------------------------------------------|
| ignatiuslumo@gmail.com      | john doe    | main   | 2     | No    | 2/14/2026 | <a href="#">Make Admin</a> <a href="#">Edit</a> <a href="#">Delete</a>   |
| 2205647@students.kcau.ac.ke | wizard wizo | main   | 27    | Yes   | 2/13/2026 | <a href="#">Remove Admin</a> <a href="#">Edit</a> <a href="#">Delete</a> |

Prev 1/1 Next

#### Knowledge Base Management

Upload official documents (PDF, DOCX, TXT) to the Qdrant vector database. The AI will immediately be able to use this information to answer user queries.

[Choose File](#) No file chosen [Upload & Ingest](#)

Supported formats: PDF, DOCX, TXT. Max size: 10MB.

### Dashboard Sections:

| Section             | Description                                                                        |
|---------------------|------------------------------------------------------------------------------------|
| Overview Statistics | Total users, active users, total chats, total messages, average session duration   |
| Activity Over Time  | Charts of daily chat/message volumes (7, 14, 30-day views)                         |
| Top Chat Topics     | Most frequently discussed topics ranked by message count                           |
| System Health       | Real-time status of Qdrant, LLM, and vector collection                             |
| User Management     | Table of all registered users with actions: Make Admin, Remove Admin, Edit, Delete |
| Knowledge Base      | Upload and ingest new documents into the AI's knowledge base                       |

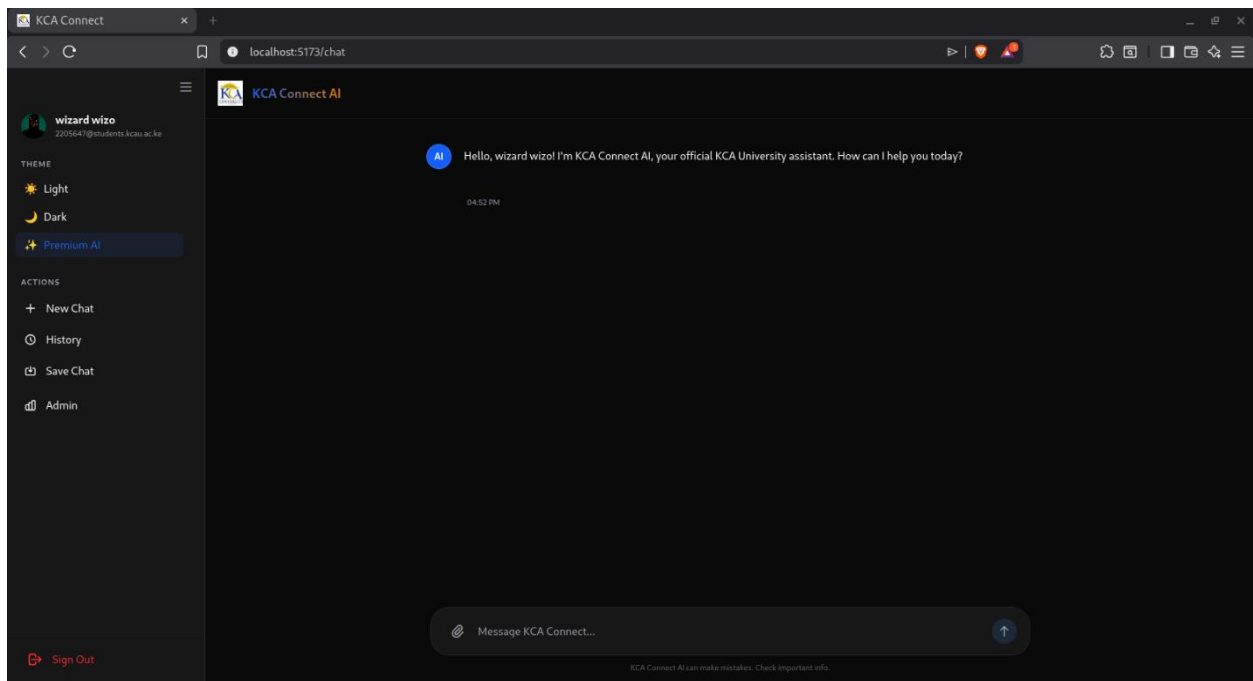
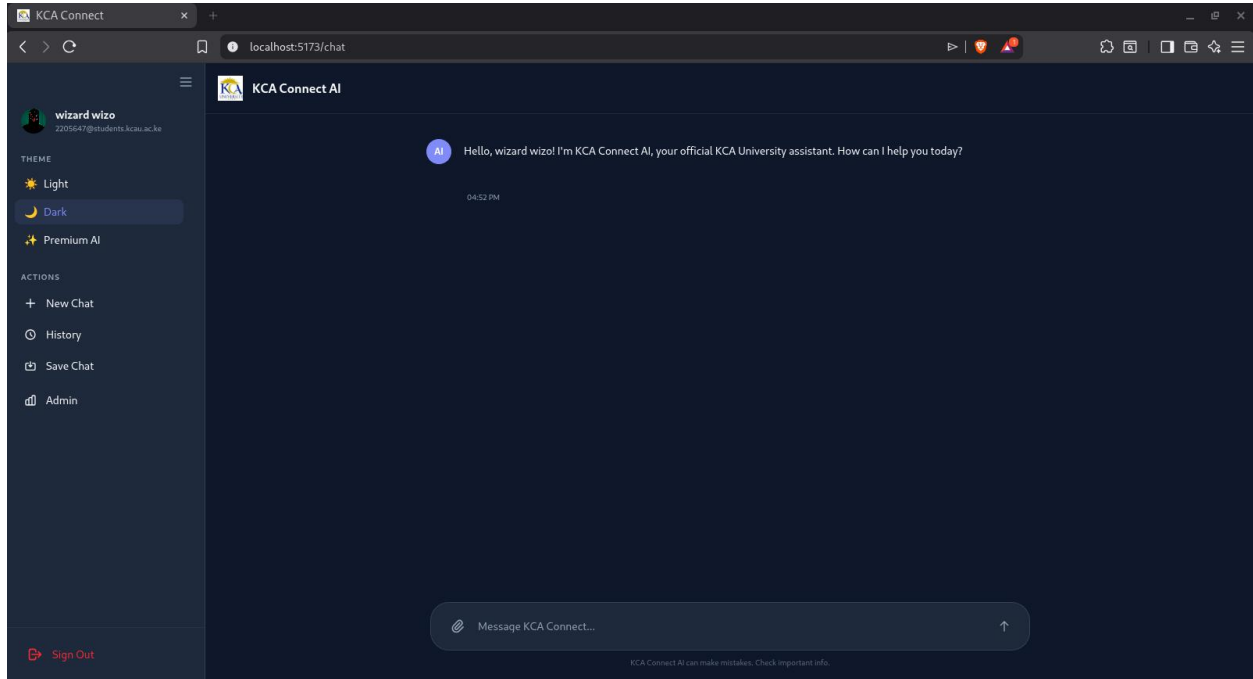
### Uploading a new document to the knowledge base:

1. Scroll to the Knowledge Base section.
2. Click "Choose File" and select a PDF, DOCX, or TXT file ( $\leq 10$  MB).
3. Click "Upload & Ingest".
4. Wait for the success message — the document is now available for AI queries.

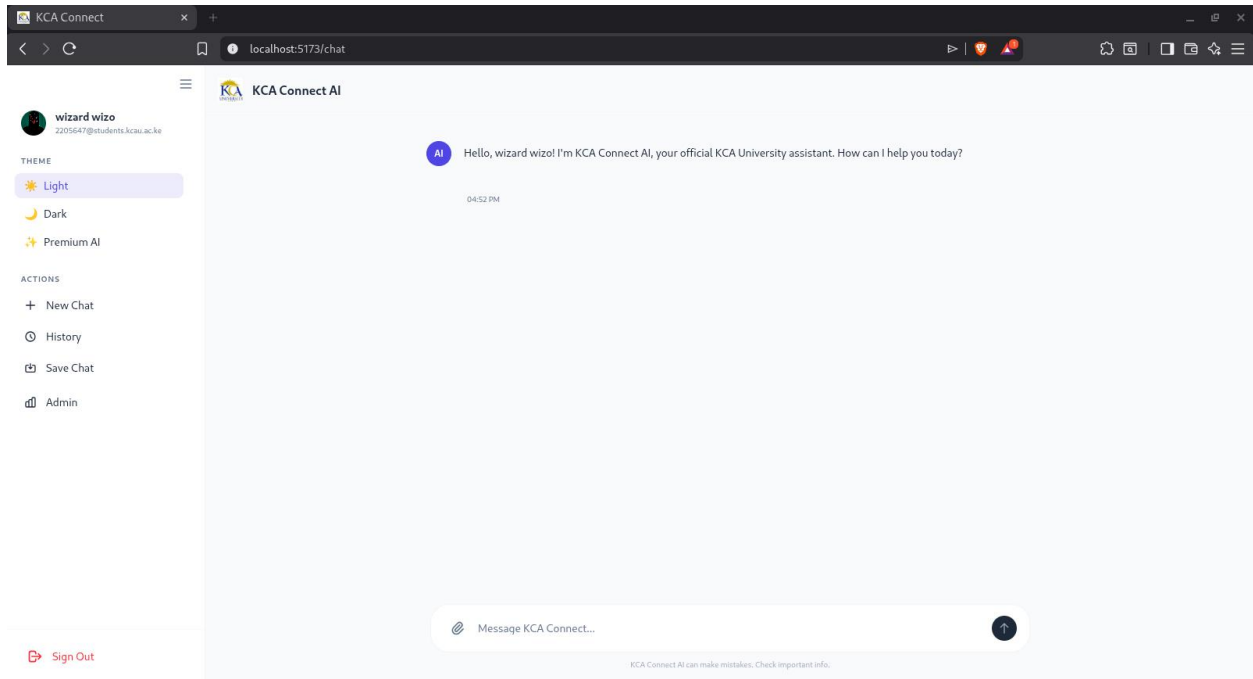
## 4.2. Customization

### Theme Settings

Three themes are available:







| Theme   | Description                                                             |
|---------|-------------------------------------------------------------------------|
| Light   | Clean white background with blue accents; ideal for bright environments |
| Dark    | Dark navy/gray background; reduces eye strain in low-light settings     |
| Premium | Luxurious gold gradient with glassmorphism effects; high-end aesthetic  |

How to change themes:

1. Click Settings .
2. Select your preferred theme.
3. The interface updates immediately.
4. Your selection is saved automatically and persists across sessions.

## 5. Maintenance and Updates

### *Basic Maintenance Tasks*

| Task                                           | Who   | Frequency |
|------------------------------------------------|-------|-----------|
| Check system health endpoint (`/health`)       | Admin | Daily     |
| Review Qdrant collection size and vector count | Admin | Weekly    |
| Monitor backend logs for errors                | Admin | Weekly    |
| Clear old/stale chat records from Supabase     | Admin | Monthly   |

### *Update Instructions*

To update the application:

10. Pull the latest code from the repository:

```
git pull origin main
```

11. Rebuild Docker containers:

```
docker-compose down
docker-compose up --build
```

12. Re-ingest documents if the knowledge base schema has changed:

```
cd backend
python scripts/ingest.py
```

### *Adding Documents to the Knowledge Base*

Via Admin Dashboard (recommended):

1. Sign in as an admin.
2. Go to Admin Dashboard → Knowledge Base section.
3. Upload the document and click "Upload & Ingest".

Via command line:

1. Place the document in the `pdf documents/` folder.
2. Run:

```
cd backend  
python scripts/ingest.py
```

### ***Backup Procedures***

1. Database Backup (Supabase): Use the Supabase dashboard to export your PostgreSQL database tables (`profiles`, `chats`).
2. Vector Store Backup: Use the Qdrant Cloud dashboard to create a snapshot of the `kca\_documents` collection.
3. Environment Backup: Keep a secure copy of your `.env` file — this contains all critical API keys.

## 6. Troubleshooting

### ***Solutions for Common Issues***

| <b>Problem</b>                 | <b>Likely Cause</b>                         | <b>Solution</b>                                                                    |
|--------------------------------|---------------------------------------------|------------------------------------------------------------------------------------|
| "Invalid credentials" on login | Wrong email/password or unconfirmed account | Reset password; check confirmation email                                           |
| AI not responding              | Backend not running; API key expired        | Check backend status at `/health`; renew API keys                                  |
| File upload failed             | File too large or unsupported format        | Ensure file is $\leq 10$ MB and is PDF/DOCX/TXT                                    |
| Chat history not saving        | Backend database unreachable; not signed in | Verify Supabase connection; sign in again                                          |
| Theme not changing             | Browser cache issue                         | Hard refresh: `Ctrl+Shift+R` (Win/Linux) or `Cmd+Shift+R` (Mac)                    |
| Slow responses                 | LLM API latency; network issues             | Check internet connection; the system will auto-fallback to next LLM               |
| Cannot access Admin Dashboard  | Not an admin account                        | Contact an existing admin to grant admin privileges                                |
| Qdrant connection error        | Docker Qdrant container stopped             | Restart with `docker-compose up -d qdrant`                                         |
| Backend won't start            | Missing `.env` file or invalid keys         | Ensure `.env` exists with all required keys; run `pip install -r requirements.txt` |

### ***Error Messages Explained***

| <b>Error Message</b>                        | <b>Meaning</b>                     | <b>Action</b>                      |
|---------------------------------------------|------------------------------------|------------------------------------|
| `"Connection refused at localhost:6333"`    | Qdrant is not running              | Start Docker services              |
| `"API key not valid"`                       | Expired or incorrect API key       | Update the key in `.env`           |
| `"Collection kca_documents does not exist"` | Documents not ingested             | Run `python scripts/ingest.py`     |
| `"Email rate limit exceeded"`               | Too many auth requests to Supabase | Wait a few minutes before retrying |

## 7. FAQ (Frequently Asked Questions)

### **General**

Q: Is KCA Connect AI free to use?

A: Yes. It is free for all KCA University students, faculty, and staff.

Q: Does it work on mobile phones?

A: Yes. The interface is fully responsive and works on smartphones and tablets in any modern browser.

Q: How accurate are the AI answers?

A: The system is trained on official KCA University documents, making answers highly accurate for topics covered. Always verify critical details (exam dates, fees) with official channels in case documents have been updated.

Q: Can I use it in Swahili?

A: Basic Swahili queries are understood. Full Swahili support is planned for future iterations.

Q: Does the system work offline?

A: No. An active internet connection is required for AI responses and database access.

### **Privacy and Security**

Q: Are my conversations private?

A: Yes. Conversations are stored securely in Supabase and are only visible to you and designated administrators (for moderation purposes). No individual chat content is shared publicly.

Q: Who can see my personal data?

A: Only you and administrators with service-level database access. Admins can see aggregate statistics, not individual message content by default.

Q: Can I delete my account?

A: Yes. Contact an administrator to request full account deletion, including all associated data.

Q: Is my data encrypted?

A: Yes. All data in transit is encrypted via HTTPS (for production). Supabase uses row-level security policies to protect user data.

### ***Features***

Q: How many files can I attach per message?

A: One file per message. You can send multiple messages with different files.

Q: Can I edit a message I've already sent?

A: No. Start a new message if you wish to rephrase a query.

Q: Can I export my chat history?

A: Currently, messages can be copied to clipboard. A full export feature is planned for a future release.

Q: What browsers are supported?

A: Chrome, Firefox, Edge, and Safari (latest versions). Google Chrome is recommended for the best experience.

## 8. Glossary and Index

### **Glossary**

| <b>Term</b>                             | <b>Definition</b>                                                                                                           |
|-----------------------------------------|-----------------------------------------------------------------------------------------------------------------------------|
| AI (Artificial Intelligence)            | Computer systems that simulate human intelligence, used here for natural language understanding and generation              |
| API (Application Programming Interface) | A set of rules and endpoints allowing software components to communicate with each other                                    |
| Cerebras                                | A third-party LLM provider used as an alternative AI backend                                                                |
| Embedding                               | A numerical vector representation of text used for semantic search                                                          |
| FastAPI                                 | A high-performance Python web framework used for the KCA Connect backend                                                    |
| Gemini                                  | Google's large language model used for generating AI responses                                                              |
| Glassmorphism                           | A modern UI design style using frosted-glass effects and translucency                                                       |
| Groq                                    | A high-speed LLM inference provider used as the default AI backend                                                          |
| HuggingFace                             | An open-source AI platform providing the `all-MiniLM-L6-v2` embedding model                                                 |
| LangChain                               | A Python framework for building LLM-powered applications, used in the RAG pipeline                                          |
| LLM (Large Language Model)              | An AI model trained on large datasets to understand and generate human language                                             |
| Markdown                                | A lightweight text formatting syntax used in AI responses (bold, lists, headers, etc.)                                      |
| Qdrant                                  | A high-performance vector database used to store and search document embeddings                                             |
| RAG (Retrieval-Augmented Generation)    | A technique combining information retrieval from a knowledge base with LLM generation for accurate, context-aware responses |
| React                                   | A JavaScript library used to build the KCA Connect frontend user interface                                                  |



|                               |                                                                                                                   |
|-------------------------------|-------------------------------------------------------------------------------------------------------------------|
| Streaming                     | Word-by-word delivery of AI responses in real time, creating a typing effect                                      |
| Supabase                      | An open-source backend-as-a-service platform providing authentication and PostgreSQL database                     |
| UAT (User Acceptance Testing) | Testing conducted with real users to validate the system meets their requirements                                 |
| Vite                          | A fast JavaScript build tool used in the KCA Connect frontend                                                     |
| Vector Search                 | Searching for semantically similar text using numerical vector representations rather than exact keyword matching |

## *Index*

| Topic                      | Section    |
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|                     |     |
|---------------------|-----|
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| Troubleshooting     | 8.6 |
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Final Year Project II